Pass it on: mesh networks

IoT · Level 4 · Lesson 2



Today you will

- Explain how a mesh passes messages hop by hop to a gateway.
- Find the weak points in a network and fix them with relays.
- Say why a mesh keeps working when one device fails.

Hop by hop

- Messages hop from neighbour to neighbour
- Another route when a device fails
- Relays fill gaps and help lonely sensors
- Protect the gateway: every route ends there

Paper mesh

1. Draw a map with ten sensors and a gateway.
2. Cut a string to the radio's range.
3. Link every pair the string reaches.
4. Find the cut-off and one-link sensors.
5. Add relays, then count the hops.

Use scissors carefully. If you try a human mesh outside, stay inside the school grounds and away from roads, and pass notes only to someone within arm's reach.

Let’s discuss

1. What happened to your network when you removed one sensor?
2. Why do sensors near the gateway run down first?
3. Where in your town would a mesh help, and what could break it?

Check yourself

- Why does a mesh keep working when one sensor fails?